Tables

Auth

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register as Organiser or Participant	None (no auth required)	firstName, lastName, dateOfBirth, email, password, roleName	201 Created / 400 Bad Request – invalid registration details / 409 Conflict – Email already exists
POST	/api/auth/login	Authenticate and start session	None (no auth required)	email, password	200 OK / 400 Bad Request / 401 Unauthorized
POST	/api/auth/logout	End session	Any (either auth required)		200 OK / 401 Unauthorized

Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	View own profile	Any (either auth required)		200 OK / 401 Unauthorized
PUT	/api/users/me	Update own profile	Any (either auth required)	firstName, lastName, phoneNumber	200 OK / 400 Bad Request / 401 Unauthorized
POST	/api/users/me/profile-picture	Upload or replace profile picture	Any (either auth required)	image file (JPG, PNG or WEBP, max 5 MB)	200 OK / 400 Bad Request / 401 Unauthorized

Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	List all events, filterable through the query parameters eventTypeID, location, fromDate, toDate and upcomingOnly	None (no auth required)		200 OK / 400 Bad Request – invalid filter values
GET	/api/events/{id}	View event detail	None (no auth required)		200 OK / 404 Not Found
GET	/api/events/mine	List events created by the logged-in Organiser, with the total enrolments for each event	Organiser		200 OK / 401 Unauthorized / 403 Forbidden
POST	/api/events	Create a new event	Organiser	eventName, description, eventDate, location, distance, eventTypeID	201 Created / 400 Bad Request – invalid details or event date not after today / 401 Unauthorized / 403 Forbidden
PUT	/api/events/{id}	Update own upcoming event	Organiser	eventName, description, eventDate, location, distance, eventTypeID	200 OK / 400 Bad Request – invalid details or event date not after today / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – event has already taken place / 409 Conflict – change would leave enrolments outside their category's age range / 409 Conflict – new distance shorter than one of its categories
DELETE	/api/events/{id}	Delete own upcoming event, along with its categories and enrolments	Organiser		204 No Content / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – event has already taken place
POST	/api/events/{id}/banner	Upload or replace event banner image	Organiser	image file (JPG, PNG or WEBP, max 5 MB)	200 OK / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
GET	/api/event-types	List all event types (Run, Walk, Cycle)	None (no auth required)		200 OK

Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/categories	List categories for an event	None (no auth required)		200 OK / 404 Not Found
POST	/api/events/{id}/categories	Add a category to an upcoming event	Organiser	categoryName, minAge, maxAge, distance	201 Created / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – event has already taken place
PUT	/api/categories/{id}	Update a category of an upcoming event	Organiser	categoryName, minAge, maxAge, distance	200 OK / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – event has already taken place / 409 Conflict – change would leave enrolments outside their category's age range
DELETE	/api/categories/{id}	Delete a category of an upcoming event, along with the enrolments that selected it	Organiser		204 No Content / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – event has already taken place

Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{id}/enrolments	Enter an upcoming event by selecting a category the participant is eligible for	Participant	categoryID	201 Created / 400 Bad Request – category not part of the event, age outside the category range, or event already taken place / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – already enrolled in this event
DELETE	/api/enrolments/{id}	Withdraw own enrolment from an upcoming event	Participant		204 No Content / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – event has already taken place
GET	/api/enrolments/me	View own enrolments	Participant		200 OK / 401 Unauthorized / 403 Forbidden
GET	/api/events/{id}/enrolments	View all enrolments for own event, optionally filtered through the query parameter categoryID	Organiser		200 OK / 401 Unauthorized / 403 Forbidden / 404 Not Found
PUT	/api/enrolments/{id}/status	Update enrolment status (e.g. Confirmed, Pending)	Organiser	status (Pending or Confirmed)	200 OK / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – result already captured / 409 Conflict – would leave fewer confirmed participants than the highest captured finishing position

Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{id}/results	Capture a result for a confirmed enrolment once the event has taken place	Organiser	finishTime, finishingPosition	201 Created / 400 Bad Request – event not yet taken place, enrolment not confirmed, or finishing position higher than the number of confirmed participants / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – result already captured
GET	/api/events/{id}/results	View all results for own event	Organiser		200 OK / 401 Unauthorized / 403 Forbidden / 404 Not Found
GET	/api/results/me	View own results, including event, category, finish time, finishing position and total finishers	Participant		200 OK / 401 Unauthorized / 403 Forbidden

Business Rules

- An event is considered to have taken place from its event date onwards, measured in South African time. Race day itself counts as having taken place.
- Events can only be created or updated with an event date after today, measured in South African time.
- Once an event has taken place, the event and its categories can no longer be updated, added to or deleted. Banner images may still be uploaded.
- Deleting an upcoming event also deletes its categories and enrolments. Deleting a category of an upcoming event also deletes the enrolments that selected it. The front end lists the affected enrolments and asks the user to confirm before the request is sent.
- Participants can only enrol in upcoming events, in a category whose age range, when it has one, includes their age on the event date, and only once per event.
- Participants can only withdraw from an enrolment while the event is upcoming. Enrolments for events that have taken place are kept as historical records.
- Results can only be captured once the event has taken place, only for Confirmed enrolments, and only once per enrolment. An enrolment's status cannot be changed after its result has been captured.
- A finishing position cannot be higher than the number of Confirmed enrolments for the event, and a Confirmed enrolment cannot be set back to Pending if that would leave fewer Confirmed enrolments than the highest finishing position already captured.
- Date of birth is required at registration for all users and cannot be changed through the profile.
- Registration only accepts the role names Organiser and Participant.
- Users must be younger than 100 at registration. Organisers must be at least 18 years old and Participants at least 10, with age measured on the registration date in South African time.
- A category must have an age range, a distance, or both.
- When an age range is set, ages must fall between 10 and 99.
- A maximum age requires a minimum age, and the maximum age cannot be lower than the minimum age.
- Categories with no age range are open to every participant.
- A category’s distance must be between 0.01 and 9999.99 km and cannot be longer than its event’s distance.
- An event’s distance cannot be reduced below the distance of any of its categories.
- A category’s age range or an event’s date cannot be changed if any existing enrolment would fall outside its category’s age range on the event date.
- Profile pictures and banner images must be JPG, PNG or WEBP files no larger than 5 MB.